Chapter Notes: Materials Around Us

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Introduction

Have you ever wondered what the world around you is made of?

Everything we see, touch, and use in our daily lives is made from different kinds of **materials**. The spoon you eat with, the clothes you wear, and even your school bag are crafted from specific materials chosen for particular reasons. Some materials are **hard** (for example, many metals), while some are **soft** (for example, cotton). Some can **bend** easily (for example, rubber), whereas others are **stiff** and can break easily (for example, glass). Learning about different materials helps us understand why things are manufactured the way they are. For instance, this scientific knowledge explains why windows are made of transparent glass and not opaque wood.

In this chapter, we will observe various objects around us, learn how to group materials systematically, study their important physical properties, and understand fundamental scientific concepts such as mass, volume, and matter.

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Observing Objects Around Us

Objects in our daily life show a remarkable variety. We constantly interact with chairs, bullock carts, bicycles, kitchen utensils, books, clothes, toys, water, stones, and much more.

Objects vary significantly in their shape, colour, size, and intended purpose. Some are roughly round, such as rubber balls and footballs, while some are square or cylindrical, such as cardboard boxes and pipes.

We can easily describe and identify objects by observing their shape, colour, or the specific material from which they are made.

Every object is made of one or more **materials**. Common everyday examples of materials include glass, metal, plastic, wood, and cloth.



Grouping objects by the material they are made of helps us to study them more easily. For example, buckets, lunch boxes, and water containers can all be grouped together as

plastic objects.

Objects can be grouped in many different ways: by their shape, by their daily use, or by the material they consist of. This systematic process of grouping objects is called **classification**.



MULTIPLE CHOICE QUESTION

Try yourself: Which property of glass makes it suitable for windows?

A Transparency

B Flexibility

C Hardness

D Roughness

View Solution

The Early Pottery in India



The oldest pottery discovered in the Indian subcontinent is estimated to be around **7,000 to 8,000 years old**. These incredible historical artifacts were excavated in places such as the **Ganga plains** (Lahuradewa) and **Baluchistan** (Mehrgarh).

From about **4000 BCE** onwards, the ancient **Sindhu-Sarasvati** culture developed advanced methods for producing **wheel-turned pottery**. These pots featured colourful coatings, known as slips, and intricate decorative painting.

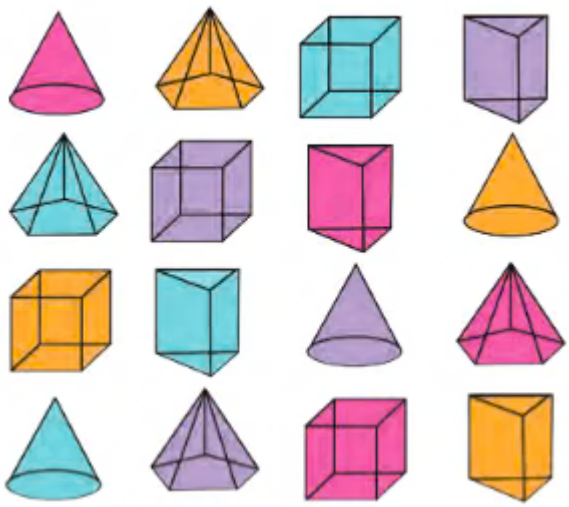
During the Harappan period (spanning about **2600 BCE to 1900 BCE**), pottery making techniques were further refined. Artisans produced bright red surfaces adorned with beautiful black designs showing **geometric patterns** and animal motifs.

The specific clay used for crafting **pots**, **dishes**, **bowls**, and other household items was chosen very carefully. It was thoroughly cleaned, sieved, kneaded, shaped perfectly on a potter's wheel, and then **baked** in hot kilns. Such baked clay items are commonly referred to as **terracotta**.

These pots were used for many vital daily functions, such as cooking and securely storing food items like grains, oil, and ghee. Today, some of these remarkable archaeological pots are proudly displayed at the **National Museum, New Delhi**.

Classification of Materials

Grouping materials helps us understand and compare their properties effectively. When objects made from the exact same material are grouped together, we can easily observe their shared properties and determine their most suitable uses. The process of classification brings to light both the **similarities** and the **differences** between various objects.



Classification is the scientific method of sorting objects into distinct groups based on their common properties.

An object may be composed of a single material or a complex combination of several materials. For example, a modern pen consists of plastic, metal components, and liquid ink.

Materials can be classified by their physical appearance (for example, whether they are **lustrous** or **non-lustrous**), their texture (for example, whether they are **hard** or **soft**), their ability to transmit light (whether they are **transparent**, **translucent**, or **opaque**), and various other measurable properties.

Knowing the specific properties of materials allows us to choose the most appropriate material for a specific use. For example, we select metals for cooking vessels because they conduct heat well, cotton for clothes because it is soft and breathable, and glass for windows because it allows natural light to enter our rooms.

Examples of Grouping

Shape: Round objects like a ball; square objects like a box; cylindrical objects like a tube.

Colour: Red objects like an apple; green objects like a leaf; blue objects like the sky.

Hardness: Soft materials can be compressed or scratched much more easily than hard materials.

Shine: Materials with naturally shiny surfaces are scientifically termed **lustrous**. A large number of metals fall into this particular category.

What are the Different Properties of Materials?

Observe and Identify Appearance of Materials



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Materials differ significantly in how they look to the naked eye. For example, a piece of unpolished wood looks entirely different from a piece of iron. Metals such as iron, copper, and aluminium have characteristic appearances that clearly distinguish them from non-metals such as wood, rubber, or plastic.

Lustrous materials have shiny, reflective surfaces. Classic examples include iron, copper, zinc, aluminium, and gold. However, it is important to note that some metals may lose their characteristic shine and become dull when exposed to air and moisture over a long period.

Non-lustrous materials do not appear shiny and generally possess a dull appearance. Common examples include paper, wood, rubber, and jute.

Always keep in mind that not all shiny objects are made of metal. The act of polishing a surface or applying thin artificial coatings of plastic or wax can make non-metals appear extremely shiny. This is why we must always remember the famous proverb: **"All that glitters is not gold."**

Which Materials are Hard?



Hardness is a property that indicates how difficult it is to physically compress or scratch a given material.

Some everyday materials, such as wax, chalk, soap, and soft wood, can be scratched with a metal key. Such materials are considered relatively soft when compared to much harder substances.

Materials that can be easily compressed or scratched are classified as **soft**, while those that require immense effort to compress or scratch are classified as **hard**.

Hardness is a comparative property. For example, rubber is tougher than a sponge but remains softer than solid iron. Thus, materials may be classified as soft or hard depending entirely on what they are being compared with.

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Let's Revise

Q: What is the difference between lustrous and non-lustrous materials?

Ans: Lustrous materials possess a natural shine, like most metals, while non-lustrous materials such as wood or paper have a dull appearance and completely lack shine.

Q: How can we tell if a material is hard or soft?

Ans: Hard materials are extremely difficult to compress or scratch, while soft materials can be easily pressed, bent, or scratched.

Materials and Light: Transparent, Translucent and Opaque





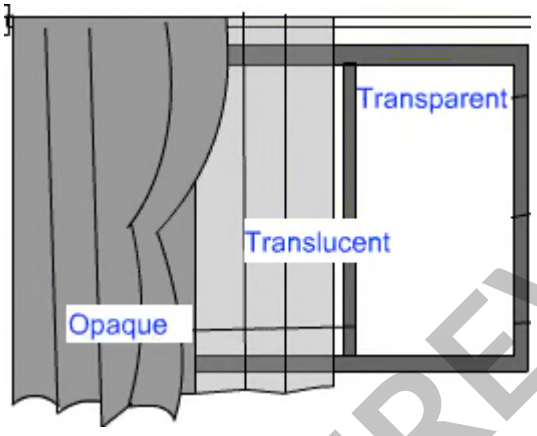
Imagine Pranav, Shreya, and Sanya are playing a fun game of **hide and seek**! Pranav hides behind a solid brick wall, Shreya hides behind the trunk of a big tree, and Sanya hides behind a frosted glass door. Meanwhile, Shreya's younger brother is carefully watching them through a clear glass window. But wait! Why can he see some things **clearly** and not others? This interesting phenomenon happens because different materials let **light** pass through them in very different ways! Let's explore these three categories:

Transparent materials allow almost all light to pass completely through them, meaning we can see objects very clearly through them. Common examples include clear glass, clean water, air, and clear cellophane paper.

Translucent materials allow only some light to pass through but do not give a crisp or clear view of the objects hidden behind them. Objects appear blurry or fuzzy. Common examples include butter paper, tissue paper, and frosted glass.

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Opaque materials completely block out light and do not allow any light to pass through. We cannot see through them at all. Common examples include solid wood, thick cardboard, and metal sheets.

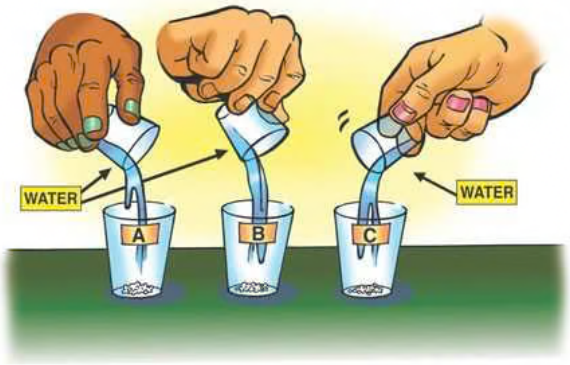


Let's Revise

Q: What is the difference between opaque and translucent materials?

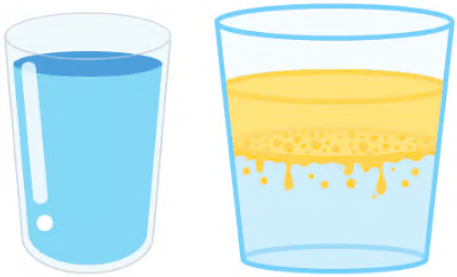
Ans: Opaque materials block all light completely, preventing you from seeing through them, while translucent materials allow some light to pass through but do not give a clear or distinct view.

Solubility: What Dissolves in Water?



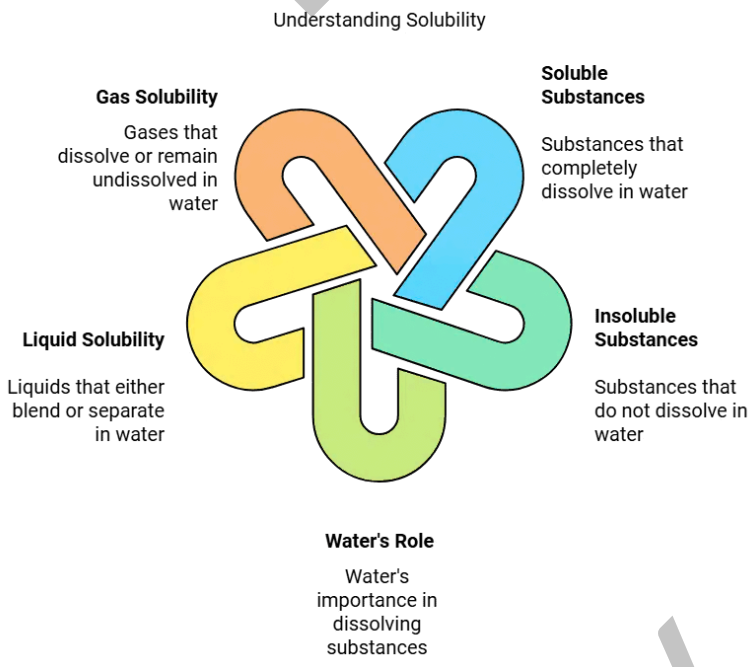
When a solid substance mixes uniformly with water and seems to completely disappear into the liquid, we say that it is **soluble** in water. Excellent everyday examples of soluble substances include sugar and salt, which dissolve effortlessly in water. On the other hand, some substances simply do not mix with water and remain clearly visible no matter how vigorously we stir them; these are scientifically known as being **insoluble** in water. Common examples of insoluble substances include sand, soil, and chalk powder.

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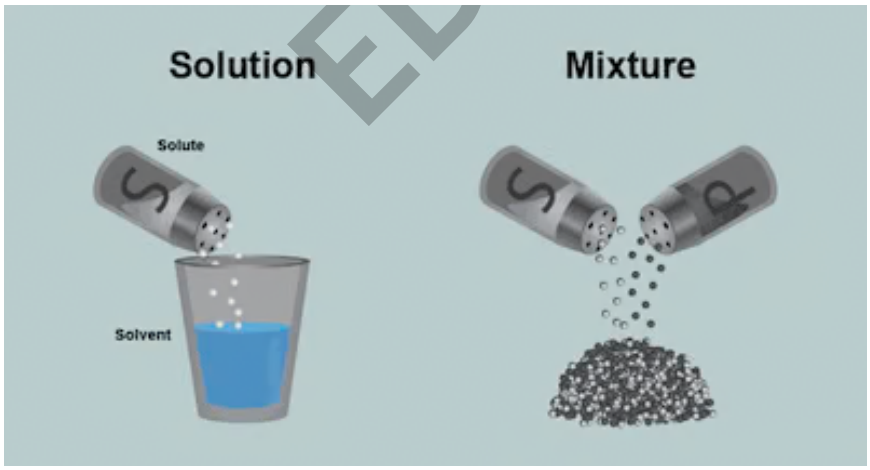
More about Solubility and Mixtures

Some liquids mix completely with water (for example, a sugar solution or lemon juice), while other liquids simply do not mix and end up forming completely separate visible layers (for example, mustard oil floats on top of water because it is less dense).
Water is incredibly crucial for the proper functioning of our bodies primarily due to its unique ability to dissolve numerous essential substances, nutrients, and minerals.
Gases also differ greatly in their solubility in water. For instance, oxygen gas dissolves in water and is absolutely essential for aquatic life. This naturally dissolved gas is highly important for the survival of fish and underwater aquatic plants.
When two or more substances mix together without any internal chemical change occurring, and they can be separated again by simple physical means, the resulting combination is called a **mixture**.



Mixing of Liquids and Solubility of Gases in Water

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Liquids that mix completely with water form uniform, homogeneous mixtures (often called solutions).

Liquids that absolutely do not mix with water form heterogeneous mixtures where distinct separated layers or floating droplets are clearly visible.

Dissolved oxygen in water is absolutely vital for fish and other aquatic organisms to breathe. Similarly, completely submerged plants living in water also actively utilise this dissolved oxygen for their vital respiration process.

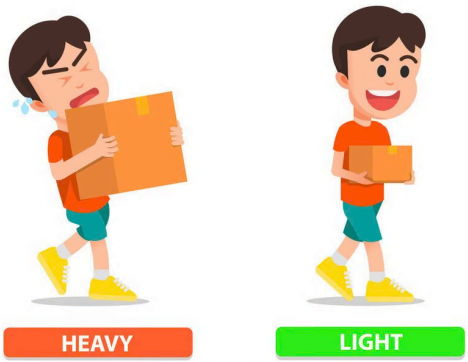
Let's Revise

Q: Why is oxygen gas important in water?

Ans: Oxygen gas naturally dissolved in water is essential because it helps fish, aquatic insects, and other aquatic life breathe and survive underwater.

Mass, Space, and Matter

How Heavy or Light?



Mass is a physical quantity that tells us exactly how much matter is present inside an object. Objects possessing more mass are heavier, whereas those with less mass are lighter.

Mass is commonly and accurately measured in standard units such as grams (g) and kilograms (kg).

Space and Volume



Volume is scientifically defined as the total amount of space taken up or occupied by an object.

The volume of liquids is most commonly measured in everyday units like litres (L) and millilitres (mL).

MULTIPLE CHOICE QUESTION

Try yourself: Which of the following materials is considered lustrous?

A Wood

B Iron

C Rubber

D Paper

View Solution

What is Matter?

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Matter is defined broadly as anything that occupies space and has mass.

All the physical materials in the universe are fundamentally made of matter and possess two incredibly important properties: **mass** and **volume**.

Examples of matter are everywhere around us and include water, sand, pebbles, your textbook, and a cup.

Mass is measured mathematically in **grams (g)** or **kilograms (kg)**.

Volume is measured mathematically in **litres (L)** or **millilitres (mL)**. The International System of Units (SI) unit for volume is the cubic metre, which is written as m^3.

Classifying Materials



Materials can be systematically grouped by properly observing their similarities and differences. Metals such as iron, copper, and aluminium are grouped together into one category because they share highly similar physical properties.

The scientific concept of grouping does not only apply to non-living materials but to natural rocks, plants, and animals as well. It deeply helps us to identify patterns of properties for easier study, observation, and daily use.

Typical characteristic properties of metals include being shiny (lustrous), physically strong, and acting as excellent conductors of both heat and electricity.

Units and Notation Rules

Unit of mass: The kilogram is the standard SI unit of mass. It should always be written without any extra space inside the unit symbol as **kg**. For example, we properly write $7\ \mathrm{kg}$ (and never write it as "7 kgs").

Units of volume: The litre is written simply as **L**, and the millilitre is written as **mL** (lowercase 'm', uppercase 'L'). For example, we accurately write $500\ \mathrm{mL}$.

The SI unit for volume is the cubic metre, mathematically represented as m^3. To understand the scale, $1\ \mathrm{m}^3$ holds exactly $1000\ \mathrm{L}$. Therefore, an equation like $2\ \mathrm{m}^3 = 2000\ \mathrm{L}$ perfectly demonstrates this relationship.

Always leave a proper space between the numerical value and the unit when writing (for example, $2\ \mathrm{kg}$ or $500\ \mathrm{mL}$).

Weight vs. Mass

In everyday conversational language, the word "weight" is very often used to mean mass. However, strictly speaking in the world of science, mass is the actual physical amount of matter inside an object and is always measured in kg or g. Weight, on the other hand, is the downward physical force exerted by gravity pulling on that mass. For most classroom science problems and daily life situations here on Earth, we generally use the term mass to properly describe how heavy an object is.

Ayurveda and the 20 Properties

Overview of Ayurveda's Property Classification

These properties are:

(i)	*guru* (heavy)	×	*laghu* (light in weight)
(ii)	*manda* (slow)	×	*tīkṣhṇa* (quick, fast)
(iii)	*hima* (cold)	×	*uṣhṇa* (hot)
(iv)	*snigdha* (unctous)	×	*rukṣha* (dry)
(v)	*śhlakṣhaṇa* (smooth)	×	*khara* (rough)
(vi)	*sāndra* (solid)	×	*drava* (liquid)
(vii)	*mṛidu* (soft)	×	*kaṭhina* (hard)
(viii)	*sthira* (stable)	×	*khāla* (moving, unstable)
(ix)	*sūkṣhma* (subtle, small)	×	*sthūla* (big, gross)
(x)	*viśhada* (non slimy)	×	*picchhila* (slimy)

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In ancient India, highly developed and structured systems of classification based on specific physical qualities were used to deeply describe substances and living things. Ayurveda, which is a traditional and highly respected Indian medical system, uniquely describes **20 properties** (commonly referred to in Sanskrit as *guna*). These are logically arranged as ten pairs of exact opposite qualities (such as hot and cold, or heavy and light).

These natural properties apply universally to physical matter, living beings, the surrounding environment, and the food we consume daily. The **Ashtanga Hridaya** is a remarkably important and historical Ayurvedic text that discusses these properties in great detail. Understanding such deeply observed qualities helps Ayurvedic practitioners in accurately diagnosing ailments and choosing the perfectly right treatments by identifying the specific physical effects different natural substances have on the human body.

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Summary and Key Concepts

Variety of materials: Every single object around us may be carefully crafted from one single material or a complex combination of several different materials.

Classification methods: Materials can be systematically and scientifically classified by their visual appearance (lustrous or dull), their physical texture (hard or soft), their light transparency (transparent, translucent, or opaque), their solubility in water, and various other measurable properties.

Properties dictate use: The specific observed properties of materials determine their absolute ideal use in daily life. For example, tough metals are used for cooking utensils because they withstand high heat, while soft cotton is used for making comfortable clothes.

Concept of matter: Anything that actively occupies physical space and has mass is defined scientifically as matter. Mass measures the exact amount of matter in an object, while volume measures the three-dimensional space that the object occupies.

Historical systems: Traditional Indian medical and scientific systems like Ayurveda also classified the qualities of substances meticulously, beautifully demonstrating India's exceptionally long and rich history of studying and applying material properties for both health and science.